Zurich University of Applied Sciences

Department of Life Sciences and Facility Management

Institute of Computational Life Sciences

Site Suitability Analysis for Wind Energy in Canton Thurgau: A Raster-based Multi-Criteria Approach

Lukas Buchmann

BSc Applied Digital Life Sciences

GISScience and Geodatabases HS25 AD23

18. January 2026

Corrector:

Pascal Ochsner

ZHAW Life Sciences und Facility Management
Institut für Umwelt und Natürliche Ressourcen
Grüentalstrasse 14
8820 Wädenswil

Content

1. Introduction - 2 -
1.1 Context: The Swiss Energy Transition - 2 -
1.2 Problem Statement: Challenges in Canton Thurgau - 2 -
1.3 Objectives - 2 -
2. Methodology - 3 -
2.1 Study Area - 3 -
2.2 Data Acquisition and Preprocessing - 3 -
2.3 Criteria Definition and Operationalization - 4 -
2.3.1 Exclusion Criteria (Constraints) - 4 -
2.3.2 Suitability Factors (Weighted) - 5 -
2.4 The Model (Map Algebra) - 5 -
3. Results - 5 -
3.1 Spatial Distribution of Suitability - 5 -
3.2 Statistical Analysis of Potential - 7 -
4. Discussion - 7 -
4.1 Interpretation of Results: The "Low-Wind" Reality - 7 -
4.2 Comparison against Official Planning - 8 -
4.3 Limitations: The "Soft" Constraints - 8 -
5. Conclusion - 9 -
6. References - 10 -
Appendix - 12 -

1. Introduction

1.1 Context: The Swiss Energy Transition

In response to global climate change and the Fukushima nuclear disaster, Switzerland adopted the Energy Strategy 2050, aiming to phase out nuclear power and reach net-zero emissions. A cornerstone of this strategy is the massive expansion of renewable energy capacities. While hydropower remains the backbone of the Swiss energy supply, it cannot be significantly expanded. Therefore, the strategy relies heavily on new renewable sources, particularly photovoltaics and wind energy, to close the supply gap left by nuclear decommissioning (SFOE, 2025).

Recent studies by ETH Zurich suggest that to meet the 2050 targets, Switzerland's wind energy capacity must increase approximately 80-fold compared to 2022 levels (ETH Zurich, 2025). Wind energy is considered particularly valuable for Switzerland's energy security because its production profile is stronger in the winter months, helping to mitigate the seasonal "winter gap" in domestic supply. Estimates suggest that roughly two-thirds of wind power generation occurs in the winter period, making it complementary to photovoltaic generation. Under Switzerland's long-term energy planning, the target for wind electricity production is around 4.3 TWh per year by 2050, compared with annual production of roughly 0.15 TWh in 2022 (VSE, 2025).

1.2 Problem Statement: Challenges in Canton Thurgau

Despite the national urgency, the deployment of wind energy in Switzerland faces significant spatial and social challenges. The complex topography, high population density, and strict landscape protection laws create a "regulatory density" that severely limits available sites (Jegen, 2015). Furthermore, while general public support for wind energy is high, projects frequently encounter strong local opposition because, as researchers at the Paul Scherrer Institute note, citizens desire secure electricity services but often reject the associated landscape impacts in their own vicinity, a challenge that persists despite broader acceptance of the technology (PSI, 2025).

Canton Thurgau illustrates some of these key challenges. As a relatively densely settled region on the Swiss Plateau, with extensive agricultural land and many villages, proposals for wind turbines have faced local resistance and acceptance issues among the residents (Bundesamt für Raumentwicklung, 2021). Moreover, wind conditions on the Swiss Plateau, tend to be more moderate than in high-wind regions such as the Alps which creates an additional hurdle to initiating suitable projects. (ETH Zurich, 2023).

1.3 Objectives

The objective of this project is to identify suitable locations for wind turbine development in Canton Thurgau that balance these conflicting economic, legal, and social requirements. To achieve this, a Geographic Information System (GIS)-based Multi-Criteria Analysis (MCA) was conducted. This method allows for the systematic overlay of spatially referenced criteria, such as wind speed, distance to settlements, and protected areas, to calculate a suitability index for the entire canton. This report documents the methodology, data processing, and final suitability map resulting from this analysis.

2. Methodology

The site suitability analysis was conducted using a raster-based Multi-Criteria Analysis (MCA) within the QGIS environment (Version 3.40.12 LTR). The workflow followed the standard GIS analysis stages.

2.1 Study Area

The study area covers Canton Thurgau in northeastern Switzerland. The canton is characterized by a rolling, hilly landscape with a high density of settlements and agricultural land. This specific topography necessitates a precise evaluation of slope and proximity constraints to ensure both economic viability and social acceptance of wind energy infrastructure (Amt für Raumentwicklung, 2024).

2.2 Data Acquisition and Preprocessing

To ensure high spatial accuracy, official geodata from federal and cantonal authorities were prioritized. All datasets were reprojected to the Swiss coordinate system CH1903+ / LV95 (EPSG: 2056) to minimize distortion in distance and area calculations.

Table 1: Data Sources

Dataset	Source	Layer	Description & Processing
Wind Atlas of Switzerland	Swiss Federal Office of Energy (BFE)	Mean annual wind speed at 100m height	Original 100m resolution raster resampled to 25m using bilinear interpolation to match topographic data.
DHM25 (Digital Elevation Model)	Swisstopo	Elevation	Digital elevation model used to derive slope and elevation parameters, chosen for cantonal-scale accuracy.
swissTLM3D	Swisstopo (tlm_strassen_strasse, tlm_namen_siedlungsname)	Roads and Settlements	Vector geometries rasterized using the Euclidean Distance tool to create continuous distance surfaces for buffering.
Federal Inventory of Landscapes and Natural Monuments (BLN)	Federal Office for the Environment (FOEN)	Protected Areas	Vector data used to define strict legal exclusion zones for the suitability analysis.

The model prioritizes data parsimony by excluding layers with minimal spatial impact in the Canton of Thurgau. Specific constraints for raised bogs (*Hochmoore*) and others, were omitted due to their negligible coverage, with the broader Federal Inventory of Landscapes and Natural Monuments (BLN) serving as a sufficient proxy for environmental protection. Similarly, hydrological features were not explicitly modeled, as potential conflicts are mitigated by the mandatory distance buffers from infrastructure and can be seen by overlaying a satellite map.

2.3 Criteria Definition and Operationalization

A "Hybrid MCA Approach" was adopted, distinguishing between Binary Constraints (Exclusion Zones, assigned values 0 or 1) and Weighted Suitability Factors (scored 1–10).

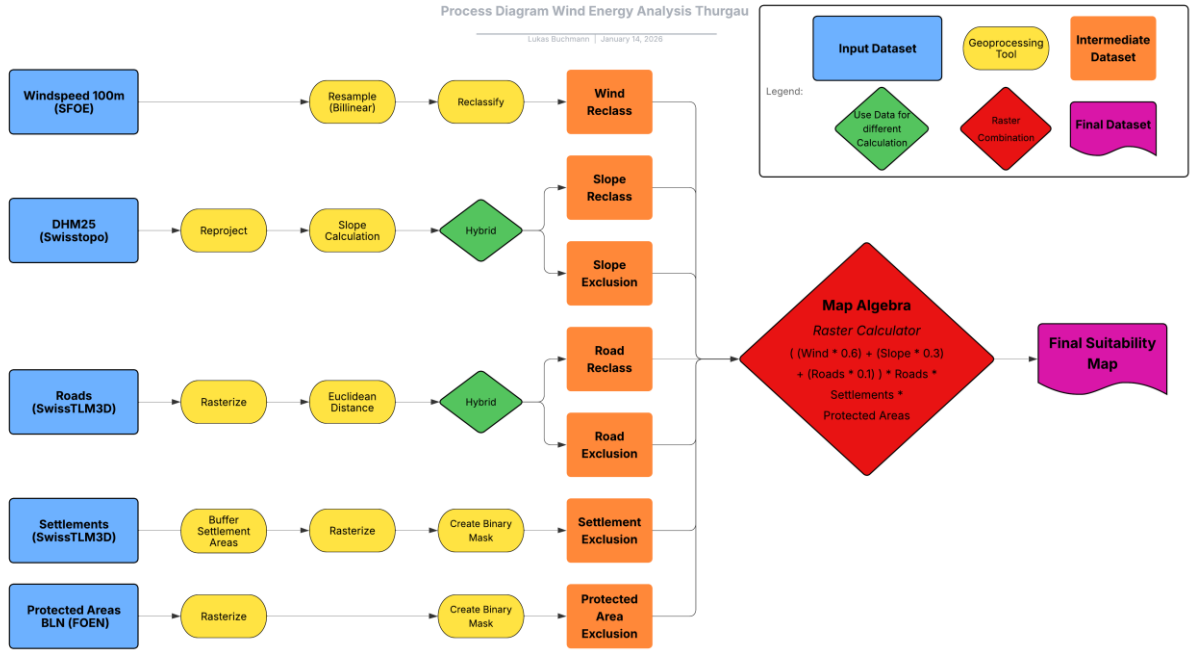

Figure 1: Process Diagram of the model, showing the Processing steps in QGIS and the final Combination.

2.3.1 Exclusion Criteria (Constraints)

Areas legally or technically unsuitable for wind turbines were assigned a value of 0 (Exclusion).

Settlement Distance (Noise Protection): To comply with the Swiss Noise Abatement Ordinance (LSV) and minimize social conflict, a strict exclusion buffer was applied. While Canton Thurgau does not mandate a fixed distance, the planning guideline suggests minimum distances dependent on noise sensitivity levels (Amt für Raumentwicklung, 2024). A conservative buffer of 300m was applied to all settlement areas to ensure compliance with LSV limits for residential zones.

Protected Areas: All zones listed in the BLN inventory were excluded (Score 0) in accordance with the Federal Act on the Protection of Nature and Cultural Heritage (NHG) (Eidgenössische Natur- und Heimatschutzkommission, 2025).

Technical Slope Limit: Areas with a slope gradient > 20% (approx. 11°) were excluded. Construction and crane access on steeper slopes are considered technically unfeasible or economically prohibitive (Bundesamt für Energie BFE, 2022).

Road Safety Buffer: To account for the risk of ice throw and structural setbacks (Baulinienabstand), a 50m safety buffer was applied around all road features.

2.3.2 Suitability Factors (Weighted)

Remaining areas were scored from 1 (Low) to 10 (High) based on their economic and logistical suitability.

Wind Speed (Weight: 60%): As the primary economic driver, wind speed received the highest weight. The reclassification values were derived from the wind speed classes defined in the Swiss Wind Atlas. Additionally, wind speeds below 4.5 m/s were defined as unsuitable and values above 6 m/s were combined as one highly suitable class. To capture variation within this range, this analysis applies a graded scoring system to differentiate site quality (BFE, 2025).

- **< 4.5 m/s:** Score 0 (Unsuitable).
- **4.5 – 5.0 m/s:** Score 3 (Marginal).
- **5.0 – 5.5 m/s:** Score 5 (Suitable).
- **5.5 – 6.0 m/s:** Score 7 (Good Suitable).
- **> 6.0 m/s:** Score 10 (Excellent).

Slope Suitability (Weight: 30%): Flatter terrain reduces civil engineering costs for foundations and crane pads. Slopes of 0–5° were assigned the highest score (10), with suitability decreasing up to the exclusion threshold of 11°.

Road Proximity (Weight: 10%): Proximity to existing roads minimizes the cost of constructing new access routes. Areas 50–500m from roads were scored highest (10), with scores decreasing for remote locations.

2.4 The Model (Map Algebra)

The final suitability map was generated using Map Algebra (Raster Calculator). The weighted linear combination of suitability factors was multiplied by the binary constraint masks to ensure that any exclusion zone resulted in a final score of 0 (Chang, 2019).

$$Final\ Score\ =\ (0.6 \cdot Wind\ +\ 0.3 \cdot Slope\ +\ 0.1 \cdot Roads) \cdot Constraints$$

3. Results

3.1 Spatial Distribution of Suitability

The MCA resulted in a suitability map covering the entire Canton of Thurgau (Figure 2). The analysis reveals a highly fragmented potential, with large contiguous suitable areas concentrated in specific geographic regions. The locations with the highest suitability scores, which are also in the cantonal structure plan of the canton Thurgau for suitable Wind energy sites, are clustered primarily along Salen-Reutenen in the north (Figure 3) and the Wellenberg/Thundorf region in the central-west (Figure 4). In contrast, the Thur valley and the densely populated Lake Constance shoreline are almost entirely classified as exclusion zones (Score 0).

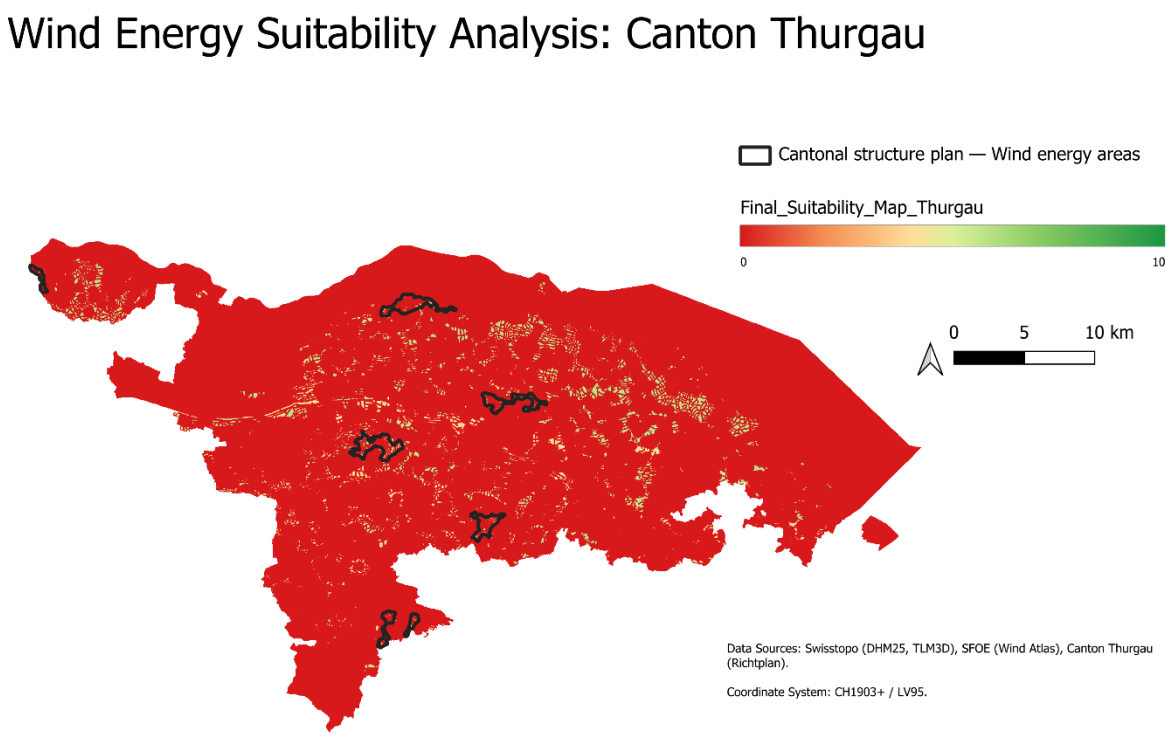

Figure 2: Spatial Distribution of Potential Wind Power Sites in Thurgau. *Suitability is classified from Low to High based on wind resource availability and logistical accessibility. Green areas indicate higher suitability. Areas in red indicate exclusion zones. The black Polygons define the political Wind energy areas designated by the canton of Thurgau.*

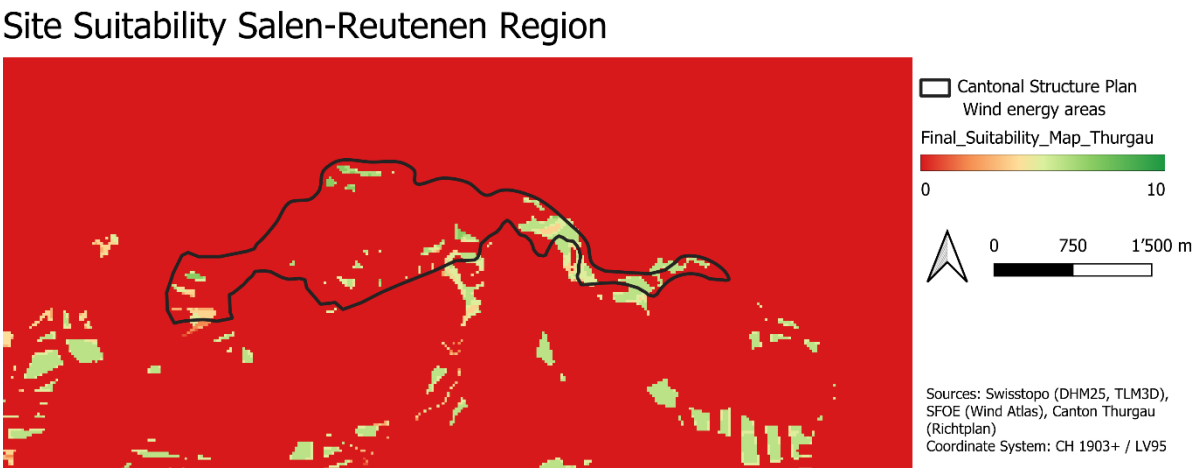

Figure 3: ***Site Suitability Analysis for Wind Energy in the Salen-Reutenen Region.*** *The map displays suitability scores (0–10) based on Swiss federal and cantonal data, with the black outline indicating official wind energy areas defined in the Cantonal Structure Plan. Green areas indicate higher suitability. Areas in red indicate exclusion zones.*

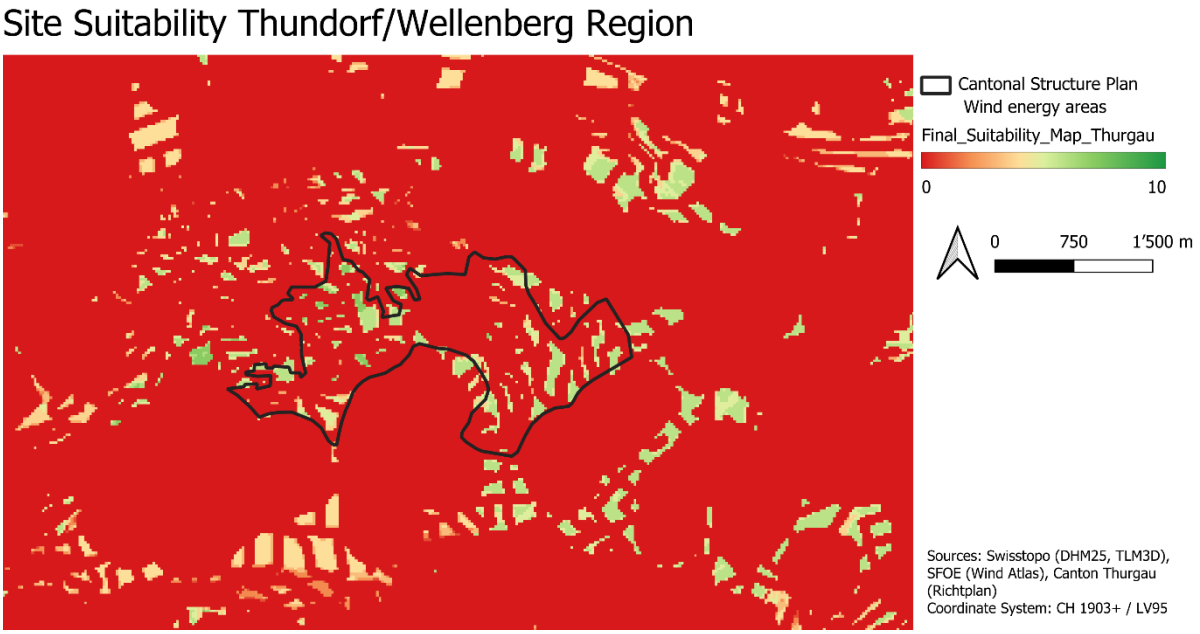

Figure 4: ***Site Suitability Analysis for Wind Energy in the Thundorf/Wellenberg Region.*** *The map displays suitability scores (0–10) based on Swiss federal and cantonal data, with the black outline indicating official wind energy areas defined in the Cantonal Structure Plan. Green areas indicate higher suitability. Areas in red indicate exclusion zones.*

3.2 Statistical Analysis of Potential

A total of 2,231 hectares (22.3 km^2) were identified as suitable for wind energy development (Score ≥ 4.4), representing approximately 2.3% of the canton's total surface area. The analysis revealed a generally moderate suitability across the region, no sites achieved 'Excellent' suitability (Score > 8). The maximum score observed was 7.0, limited to a small area of 43 hectares (0.43 km^2). The remaining 97.7% of the canton (95,774 hectares) was classified as unsuitable, either due to strict legal exclusion zones (Score 0) or insufficient wind resources.

4. Discussion

4.1 Interpretation of Results: The "Low-Wind" Reality

The analysis reveals a strong contrast between the theoretical expectations of the Energy Strategy 2050 and the physical reality in Canton Thurgau. The maximum suitability score achieved was 7.0 out of 10. This result is not an artifact of the model but a direct reflection of the local wind conditions. As a plateau canton (Mittelland), Thurgau generally experiences mean wind speeds between 4.5 and 5.5 m/s (New Energy Scout GmbH, 2014). In the weighted model (60% wind), a site with "Marginal" to "Suitable" wind (Score 3–5) cannot mathematically achieve a high total score, even if slope and infrastructure conditions are perfect. This confirms that Thurgau is a "low-wind" region where economic viability depends on maximizing other factors (infrastructure access) and utilizing modern "weak-wind" turbines (Amt für Raumentwicklung, 2024).

4.2 Comparison against Official Planning

The validity of the MCA model couldn't be tested properly. Instead, a comparison with official planning sites was made by overlaying the calculated high-suitability clusters (Score ≥ 4.4) with the six official Wind Energy Areas (Windenergiegebiete) designated in the Cantonal Structure Plan (Richtplan).

- **Agreement:** The model successfully identified many suitable locations in the official Wind Energy Areas designated by the Canton of Thurgau. This spatial coincidence confirms that the chosen criteria accurately reflect the technical planning principles used by cantonal authorities.
- **Discrepancy:** The model identified significantly more potential area (2,231 ha) than is currently designated in the Richtplan. This discrepancy highlights the difference between technical potential and political feasibility. The Richtplan zones are the result of a political negotiation process that likely excluded technically viable sites due to unmodeled social opposition, land ownership fragmentation, or landscape aesthetics.

By overlaying the 25 m resolution technically feasible areas from the model with the broader politically permissible zones, precise locations suitable for wind turbine construction can be identified.

4.3 Limitations: The "Soft" Constraints

While the model incorporates the primary physical and legal constraints, several "soft" but critical factors were excluded due to data availability and the scope of the project. These factors likely reduce the actual realizable potential significantly:

- **No Validation data:** A key limitation of the project is the absence of operational wind turbines in Canton Thurgau, which prevents direct validation of the model against actual wind energy production sites. The only available reference data are politically designated zones, which reflect permitted areas rather than realized installations. A possible solution for this problem in the future would be to validate the model on a different canton to ensure its accuracy. However, that would have exceeded the scope of this project.
- **Aviation and Radar (Skyguide/Military):** Canton Thurgau is heavily affected by aviation infrastructure. The Frauenfeld region hosts a military airfield, and large parts of the canton lie within the surveillance zones of Skyguide radar systems (e.g., Zurich/Kloten approach sectors). Wind turbines can interfere with radar signals ("clutter"), leading to project vetos. Recent federal reports indicate that aviation conflicts block a significant portion of wind projects in the Swiss Plateau (ARCS, 2025).
- **Biodiversity (Bird Migration):** The model utilized a static protected area mask (BLN). However, Thurgau lies on important migratory corridors for birds. The Swiss Ornithological Institute warns that turbines on ridges (often the windiest sites) pose collision risks for raptors like the Red Kite (*Milvus milvus*) (Schaub, 2012). Detailed Environmental Impact Assessments (UVP) would likely exclude further high-scoring areas.
- **Landscape and Visual Constraints:** Many of the locations identified as suitable in this analysis are likely to be excluded in practice due to landscape and visual impact considerations, which were not explicitly included in the model. In particular, sites

visible from Lake Constance are expected to face the strongest restrictions to preserve scenic and cultural values.

- **Infeasible Sites:** Areas covered by permanent water bodies, railways or bridges should be excluded from the analysis, as turbine construction in these locations is technically unfeasible and legally restricted.
- **Raster Resolution:** The 25m raster resolution generalizes real-world property boundaries. In practice, a single "suitable" pixel might be split between multiple landowners, making lease negotiations difficult, a non-spatial constraint that GIS cannot model.
- **Sensitivity Analysis:** A sensitivity analysis would have further validated the model's robustness by testing how variations in criteria weights influence the final output. Consequently, the results should be interpreted as a baseline scenario based on the current weights, acknowledging that different stakeholder priorities might shift the high-suitability zones.

In order to create an officially usable model for the canton of Thurgau, all factors defined by the Swiss government in relation to wind energy would have to be taken into account (Bundesamt für Energie BFE, 2019), and more intensive research would have to be carried out on different types of wind turbines and their requirements and effects.

5. Conclusion

This study used a GIS-based Multi-Criteria Analysis to identify suitable sites for wind energy in Canton Thurgau, integrating economic, technical, and legal criteria. While the canton has theoretical potential, its moderate wind conditions and the social acceptance limit economic viability. The analysis identified 2,231 ha of suitable land, exceeding areas currently designated in the cantonal structure plan. However, the maximum suitability score was only 7/10, highlighting the need to minimize costs and improve social acceptance. Model comparison against the wind energy areas defined by the canton of Thurgau show that the model identifies the specific, technically feasible positions at each area. Nonetheless, formal validation is constrained by the absence of comprehensive data on existing wind turbine installations in Thurgau. Overall, the findings suggest that the canton can contribute to Switzerland’s Energy Strategy 2050, but development strategies should prioritize a limited number of highly efficient sites rather than widespread expansion.

6. References

Amt für Raumentwicklung. (2024, November 20). *Leitfaden für die Planung von Windenergieanlagen im Kanton Thurgau*. https://raumentwicklung.tg.ch/public/upload/assets/111161/Leitfaden%20f%C3%BCr%20die%20Planung%20von%20Windenergieanlagen%20im%20Kanton%20Thurgau.pdf?fp=5

ARCS. (2025, Dezember 2). *Projekt Windturbinen | Aviation Research Center Switzerland*. https://www.arcs.aero/de/node/278

BFE, B. für E. (2025, Dezember 2). *Der Windatlas Schweiz gibt Auskunft über Jahresmittel der modellierten Windgeschwindigkeit und Windrichtung auf fünf Höhenstufen über Grund: 50, 75, 100, 125 oder 150m*. Windatlas Schweiz. http://www.windatlas.ch

Bundesamt für Energie BFE. (2019, Februar 15). *Wichtige Geodatensätze des Bundes im Zusammenhang mit Windenergie*. 9404-Liste_von_Geodaten_im Zusammenhang_mit_Windenergie_v1.1.pdf

Bundesamt für Energie BFE. (2022, August 24). *Windpotenzial Schweiz 2022*. https://www.newsd.admin.ch/newsd/message/attachments/72771.pdf

Bundesamt für Raumentwicklung. (2021, August 24). *Richtplan-kanton-thurgau-pruefungsbericht-genehmigung-uvek-vom-24082021*. https://www.are.admin.ch/dam/de/sd-web/RnwUSa2yajDo/richtplan-kanton-thurgau-pruefungsbericht-genehmigung-uvek-vom-24082021.pdf

Chang, K.-T. (2019). *Introduction to geographic information systems* (Ninth Edition). McGraw-Hill Education.

Eidgenössische Natur- und Heimatschutzkommission. (2025, Dezember 2). *Landschaften und Naturdenkmäler*. https://www.enhk.admin.ch/de/themen/landschaften-und-naturdenkmaeler

ETH Zurich. (2023, März 20). *Where should wind turbines be constructed in Switzerland?* ETH Zurich. https://ethz.ch/en/news-and-events/eth-news/news/2023/03/where-should-wind-turbines-be-constructed-in-switzerland.html

ETH Zurich. (2025, Mai 22). *Where power will come from in 2050*. ETH Zurich. https://ethz.ch/en/news-and-events/eth-news/news/2025/05/where-power-will-come-from-in-2050.html

Jegen, D. M. (2015). *Energy transition and challenges for wind energy in Switzerland*.

New Energy Scout GmbH. (2014, September 10). *Windpotentialstudie Kanton Thurgau*. https://energie.tg.ch/public/upload/assets/15746/Windpotentialstudie-TG_Teil-2_2014-09-10.pdf?fp=1

PSI. (2025, Januar 15). *"The biggest challenge is lacking public acceptance of wind turbines"* | *News & Events* | *PSI*. https://www.psi.ch/en/news/psi-stories/the-biggest-challenge-is-lacking-public-acceptance-of-wind-turbines

Schaub, M. (2012). Spatial distribution of wind turbines is crucial for the survival of red kite populations. *Biological Conservation*, *155*, 111–118. https://doi.org/10.1016/j.biocon.2012.06.021

SFOE, S. F. O. of E. (2025, Mai 22). *Energy Strategy 2050*. https://www.bfe.admin.ch/bfe/en/home/politik/energiestrategie-2050.html

VSE. (2025, Mai 22). https://www.strom.ch/de/wissen/technologien

Appendix

Declaration on the Use of Generative AI Systems

In accordance with the ZHAW guidelines (Z-RL-Richtlinie KI bei Leistungsnachweisen, 01.04.2023), I hereby declare the use of generative AI systems in the creation of this project report.

1. Tool Used: Google Gemini (Large Language Model).

2. Purpose and Extent of Use:

- **Conceptual Support:** The AI was used as a "sparring partner" to discuss the methodology of the Multi-Criteria Analysis, specifically the differentiation between exclusion and suitability criteria (Hybrid Approach).
- **Technical Troubleshooting:** The AI assisted in debugging QGIS processes, specifically regarding the "Raster Calculator" syntax, handling NoData values in rasterization, and the "Reclassify by Table" workflows.
- **Literature Search Assistance:** The AI was used to identify relevant keywords and official Swiss federal documents (e.g., "Windpotenzial Schweiz 2022", "Leitfaden Windenergie Thurgau"). All sources suggested by the AI were manually verified, read, and cited by the author.
- **Textual Support:** The AI was used to generate outlines and structure suggestions for this report. The final text and argumentation were written and verified by the author. No text passages were copied verbatim without review.

3. Author's Responsibility: I confirm that I have verified all outputs generated by the AI for accuracy and relevance. The scientific argumentation, the execution of the GIS analysis, and the conclusions drawn are my own intellectual work.